

Incoterms 2020 & Global Trade Cost Engine

Production Delivery Report • React & Vite Web Application • ICC 2020 Standards

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Live Production URL: <https://mohammed905-stack.github.io/incoterms-trade-engine/>

Source Repository: <https://github.com/mohammed905-stack/incoterms-trade-engine>

1. Executive Summary & Purpose

The Incoterms 2020 & Global Trade Cost Engine is an interactive, high-precision web application engineered to bridge complex international trade regulations with mathematical landed cost accounting.

Governed strictly by official International Chamber of Commerce (ICC) 2020 standards, the engine demystifies the eleven Incoterms rules across all transport modes (Any Mode vs. Sea & Inland Waterway), allocating risk, transport, customs clearance, and insurance responsibilities between Buyer and Seller. By integrating dynamic financial inputs (Base Commercial Invoice, Terminal Handling, Ocean/Air Freight, Marine Insurance, Customs Tariffs, and Port Clearance), the engine provides an instantaneous, zero-error audit of total landed costs and statutory compliance.

2. Core Functional Capabilities

- **11 ICC Incoterms 2020 Rules Support:**

EXW, FCA, CPT, CIP, DAP, DPU, DDP (Any Mode) and FAS, FOB, CFR, CIF (Sea/Waterway) with precise point-of-delivery and risk-transfer demarcation

- **Mathematical Landed Cost Engine:**

Dynamically calculates Ex-Works, Port Delivery, Freight, CIF/CIP Values, Customs Duties, VAT, Port Clearance Fees, and Final Landed Unit Cost

- **Interactive Responsibility Matrix:**

Granular visualization of 10 key supply chain obligations (Packaging, Export License, Carriage, Insurance, Import Formalities) color-coded by party

- **Customs & Compliance Auditing:**

Proactively alerts freight forwarders to mandatory documentation requirements, dangerous goods surcharges, and customs valuation benchmarks

- **Exportable Trade Reports:**

Allows export and verification of quote matrices for logistics coordinators, supply chain officers, and commercial banking review.

3. Modern Front-End Architecture & Technologies

Framework & Build Engine:

React 18 / 19 with Vite 6 providing sub-millisecond HMR and optimized tree-shaken static bundles.

Type Safety & Data Contracts:

Strict TypeScript with typed interfaces for Incoterms rules, cost itemizations, and validation schemas.

Styling & Responsive Layout:

Tailwind CSS utility styling with dark-mode aesthetic, cyber-logistics glassmorphism, and responsive grid layouts.

Component Modularity:

Clean separation of concerns: CostCalculator, IncotermsMatrix, RuleDetailModal, and ComplianceAuditor.

State Management:

Reactive client-side state handling with instant recalculated yields upon parameter adjustments.

Continuous Deployment:

Automated GitHub Pages workflow deploying production builds directly from git origin master branch.

4. Official ICC 2020 Standards Compliance Breakdown

- Any Transport Mode (Multimodal): EXW (Ex Works), FCA (Free Carrier), CPT (Carriage Paid To), CIP (Carriage and Insurance Paid to - Institute Cargo Clauses A by default), DAP (Delivered at Place), DPU (Delivered at Place Unloaded - replaces DAT), DDP (Delivered Duty Paid).
- Sea & Inland Waterway Specific: FAS (Free Alongside Ship), FOB (Free On Board), CFR (Cost and Freight), CIF (Cost, Insurance and Freight - Institute Cargo Clauses C minimum requirement).
- Risk Demarcation: Exact point-of-transfer demarcated visually to eliminate cross-border dispute ambiguity and prevent demurrage or detention penalties.

5. Verification Endpoints & Project Resources

Verified Live Production Deployment:

<https://mohammed905-stack.github.io/incoterms-trade-engine/>

Verified GitHub Source Repository:

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